BJT small-signal I

Answers and worked solutions · Day 10 · Mon 2 Nov · pack E6

5. Practice

From your course files

WS6 Q1 (course question).

Take V_T = 25 mV, as the solution does.

$$g_m = \frac{I_C}{V_T} = \frac{1\ \text{mA}}{0.025\ \text{V}} = \mathbf{40\ mA/V}$$

$$\alpha = \frac{\beta}{1+\beta} = \frac{100}{101}, \qquad r_e = \frac{\alpha}{g_m} = \frac{100}{101} \times \frac{1}{40\ \text{mA/V}} = \mathbf{24.8\ \Omega}$$

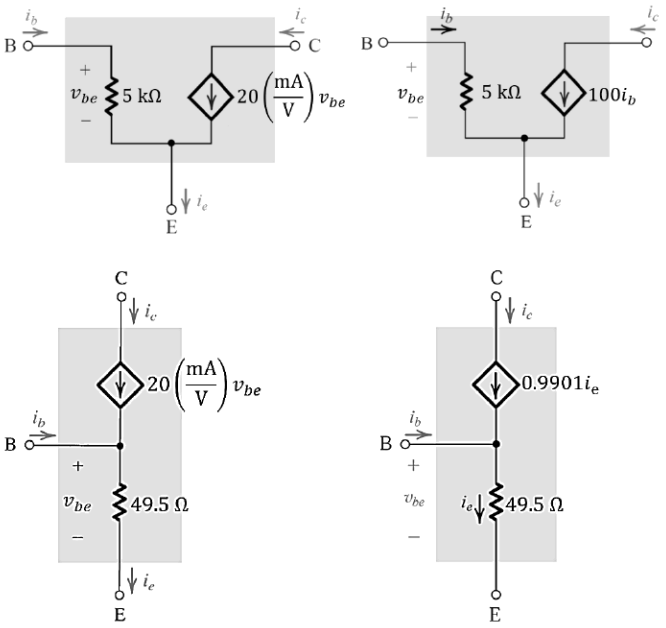
$$r_\pi = \frac{\beta}{g_m} = \frac{100}{40\ \text{mA/V}} = \mathbf{2.5\ k\Omega}$$

The solution's first line shows "0.5 mA" in the g_m fraction. That is a typo: I_C = 1 mA, and the 40 mA/V result is right.

WS6 Q2 (course question).

$$r_\pi = \frac{\beta}{g_m} = \frac{100}{20\ \text{mA/V}} = \mathbf{5\ k\Omega}, \qquad r_e = \frac{\alpha}{g_m} = \frac{100}{101 \times 20\ \text{mA/V}} = \mathbf{49.5\ \Omega}$$

The four models are hybrid-π with $g_m v_{be}$ or with βi_b, and T with $g_m v_{be}$ or with αi_e (α = 0.9901). The official drawings:



WS6 Q2: the four small-signal models from the official solution. Top: hybrid-π with $g_m v_{be}$ and with $100 i_b$. Bottom: T model with $g_m v_{be}$ and with $0.9901 i_e$.

WS6 Q4 (course question).

$\beta = \infty$, so $\alpha = 1$. In the small-signal circuit the +3 V rail is ground, so the 10 kΩ runs from the collector to ground. The base is grounded, the current source is an open circuit, and v_i drives the emitter through the (shorted) capacitor.

T model: r_e = 0.125 kΩ from the base (ground) to the emitter, and αi_e from the collector to the base, with i_e flowing from base to emitter.

- Collector: $v_c = -\alpha i_e (10\ \text{k}\Omega)$
- Emitter: $i_e = -\dfrac{v_i}{r_e}$

$$v_c = -\alpha \left(-\frac{v_i}{r_e} \right) (10\ \text{k}\Omega) \ \Rightarrow\ \frac{v_c}{v_i} = \frac{\alpha (10\ \text{k}\Omega)}{r_e} = \frac{1 \times 10\ \text{k}}{0.125\ \text{k}} = \mathbf{80}$$

Positive: a **non-inverting** amplifier. The signal enters at the emitter with the base at AC ground, so this is a common-base stage (more on CB in E7). The DC values (V_C = 1 V, I = 0.2 mA) only confirm that the transistor is active; r_e is given.

WS6 Q5 (course question).

1. The small-signal circuit. DC sources off and capacitors shorted:

- v_{sig} in series with 2 kΩ drives the base; the 10 kΩ runs from base to ground.
- Hybrid-π: r_π from base to emitter; $g_m v_{be}$ and r_o from collector to emitter.

- Z is grounded through its capacitor, so the emitter is at AC ground (the 1 mA source is open).
- At the collector: the 8 kΩ R_C (its top is AC ground) and the 8 kΩ load on Y, both to ground.

2. Gain. The output is the voltage across $r_o \parallel 8\text{k} \parallel 8\text{k}$, with $g_m v_{be}$ flowing into the collector from that network:

$$v_y = -g_m v_{be}\,(r_o \parallel 8\text{k} \parallel 8\text{k}) \qquad (1)$$

v_{be} is the voltage across $r_\pi \parallel 10\text{k}$, which forms a divider with the 2 kΩ source resistance:

$$v_{be} = v_{sig}\,\frac{r_\pi \parallel 10\text{k}}{(r_\pi \parallel 10\text{k}) + 2\text{k}} \qquad (2)$$

Numbers: $r_\pi \parallel 10\text{k}$ = 2.016 kΩ, so the divider is 0.502. $r_o \parallel 8\text{k} \parallel 8\text{k} = 101\text{k} \parallel 4\text{k}$ = 3.848 kΩ.

$$\frac{v_y}{v_{sig}} = -39.604 \times 0.502 \times 3.848 = \mathbf{-76.49\ V/V}$$

An inverting amplifier.

WS6 Q6 (course question).

Which DC method. $\beta R_E = 180(2.2\text{ k}) = 396$ kΩ is less than $10R_2 = 560$ kΩ, so the approximate method is not allowed. Use the exact (Thevenin) method.

DC (exact): $R_{Th} = 220\text{ k} \parallel 56\text{ k} = 44.64$ kΩ and $E_{Th} = \dfrac{56}{276}(20) = 4.06$ V.

$$I_B = \frac{4.06 - 0.7}{44.64\text{ k} + 181(2.2\text{ k})} = 7.58\ \mu\text{A}, \quad I_E = 181 I_B = 1.372\text{ mA}, \quad I_C = 180 I_B = 1.365\text{ mA}$$

1. $r_e = 26\text{ mV}/1.372\text{ mA} = \mathbf{18.94\ \Omega}$. The solution gives 18.92 Ω because it rounds I_B to 7.59 µA.

2. $V_B = V_{BE} + I_E R_E = 0.7 + 1.372(2.2) = \mathbf{3.72\ V}$ (the same as $E_{Th} - I_B R_{Th}$). The OCR'd solution did not show a V_C value; from the solution's own currents, $V_C = V_{CC} - I_C R_C = 20 - 1.365(4.7) = \mathbf{13.58\ V}$. Check: $V_E = 3.02$ V, so $V_{CE} = 10.6$ V. Active.

3. $\beta r_e = 180(18.94) = 3.41$ kΩ, so $Z_i = R_{Th} \parallel \beta r_e = 44.64\text{ k} \parallel 3.41\text{ k} = \mathbf{3.17\ k\Omega}$.

The figure gives the output conductance $g_{os} = 30\ \mu$S (the same quantity as h_{oe}), so $r_o = 1/g_{os} = 33.3$ kΩ. That is only 7× R_C, so keep it: $R_C \parallel r_o = 4.12$ kΩ, and

$$A_V = -\frac{R_C \parallel r_o}{r_e} = -\frac{4119}{18.94} = \mathbf{-217.4}$$

The solution, with r_e = 18.92 Ω and $R_C \parallel r_o$ rounded to 4.12 kΩ, gets −217.76. Ignoring r_o would give −248.

New questions

N1 (easy) [4 marks].

a. $g_m = 2/26 = \mathbf{76.9\ mA/V}$ and $r_\pi = 150/0.0769 = \mathbf{1.95\ k\Omega}$.

$I_E = 2(151/150) = 2.013$ mA, so $r_e = 26/2.013 = \mathbf{12.91\ \Omega}$. Also $r_o = 80/2\ \text{mA} = \mathbf{40\ k\Omega}$.

b. $(\beta+1)r_e = 151(12.91) = 1.95\ \text{k}\Omega = r_\pi$. ✓ The identity is exact because $I_E = (\beta+1)I_B$.

N2 (medium) [7 marks].

a. $I_B = (15 - 0.7)/680\ \text{k} = 21.03\ \mu$A, $I_C = 120 I_B = 2.524$ mA, and $I_E = 121 I_B = 2.545$ mA. $V_{CE} = 15 - 2.524(4.7) = \mathbf{3.14\ V}$ > 0.2 V, so the BJT is active.

b. $r_e = 26/2.545 = \mathbf{10.22\ \Omega}$ and $r_o = 100/2.524\ \text{mA} = \mathbf{39.6\ k\Omega}$.

c. $\beta r_e = 1.226$ kΩ, so $Z_i = 680\ \text{k} \parallel 1.226\ \text{k} = \mathbf{1.22\ k\Omega}$.

Without r_o: $Z_o = 4.7$ kΩ and $A_v = -4700/10.22 = \mathbf{-460}$.

With r_o: $Z_o = 4.7\ \text{k} \parallel 39.6\ \text{k} = \mathbf{4.20\ k\Omega}$ and $A_v = -4202/10.22 = \mathbf{-411}$. Here $r_o/R_C = 8.4 < 10$, so r_o should be kept.

d. $v_o = -411(5\ \text{mV})\sin\omega t = \mathbf{-2.06 \sin \omega t\ V}$. The downward headroom is $V_{CE} - 0.2 = 2.94$ V and the upward headroom is $V_{CC} - V_{CE} = 11.9$ V. A 2.06 V peak fits, so there is no clipping, though there is not much margin on the negative side. The output is inverted (180°).

N3 (medium) [8 marks].

a. $\beta R_E = 150$ kΩ ≥ $10R_2 = 100$ kΩ, so the approximate method is allowed.

$V_B = 10(12)/57 = 2.105$ V, $V_E = 1.405$ V and $I_E = 1.405$ mA. Then $V_C = 12 - 1.405(3.3) = \mathbf{7.36\ V}$ and $V_{CE} = 12 - 1.405(4.3) = \mathbf{5.96\ V}$. Active.

b. $r_e = 26/1.405 = \mathbf{18.50\ \Omega}$

c. $R' = 47\ \text{k} \parallel 10\ \text{k} = 8.25$ kΩ and $\beta r_e = 2.78$ kΩ, so $Z_i = \mathbf{2.08\ k\Omega}$.

$Z_o = 3.3\text{ k} \parallel 60\text{ k} = \mathbf{3.13\ k\Omega}$ and $A_v = -3128/18.50 = \mathbf{-169}$, with a 180° phase shift. If r_o is ignored, A_v = −178.

For comparison, the exact (Thevenin) method gives $I_E = 1.33$ mA, $r_e = 19.5\ \Omega$ and $A_v = -160$. The test only guarantees about 10 % accuracy, and here the error is about 5 %.

d. $h_{ie} = \beta r_e = 2.78$ kΩ, $h_{fe} = \beta = 150$, $h_{oe} = 1/r_o = 16.7\ \mu$S. ($h_{re} \approx 0$.)

N4 (multi-part, chains) [9 marks] — your Practical 3 amplifier.

a. $\beta R_E = 270$ kΩ ≥ 100 kΩ, so the approximate method is allowed. $V_B = 10(10)/56 = \mathbf{1.786\ V}$, $V_E = 1.086$ V and $I_E = 1.086/0.98\text{ k} = \mathbf{1.108\ mA}$. The measured values (1.735 V, 1.12 mA) are within 3 % and 1 %. The lower V_B is base current loading the divider.

b. $r_e = 26/1.12 = \mathbf{23.2\ \Omega}$ and $r_o = 100/1.12\text{ mA} \approx \mathbf{89\ k\Omega}$.

c. $R' = 46\text{ k} \parallel 10\text{ k} = 8.21$ kΩ and $\beta r_e = 275.5(23.2) = 6.40$ kΩ, so $Z_i = \mathbf{3.60\ k\Omega}$ (measured 3.51 kΩ with V_s = 50 mV).

With $r_o = \infty$: $Z_o = 3.81$ kΩ and $|A_v| = 3810/23.21 = \mathbf{164}$. With r_o: $Z_o = 3.65$ kΩ and $|A_v| = 3654/23.21 = \mathbf{157}$.

d. C_E is not a perfect short. At 1 kHz, $X_{C_E} = 1.59\ \Omega$, which adds about 7 % to r_e in the emitter. Your findings show this brings the prediction to 147. The rest is component and β tolerance. (Dividing by $r_e + X_{C_E}$ is a preview of the unbypassed analysis in E7.)

e. $R_C \parallel r_o \parallel R_L = 3.81\text{ k} \parallel 89\text{ k} \parallel 4.7\text{ k} = 2.06$ kΩ, so $|A_{vL}| = 2055/23.21 = \mathbf{88.6}$. The measured value was 77.1; with the X_{C_E} correction your findings predicted 82.9. Loading roughly halves the gain, which is why E7 treats R_L properly.

N5 (hard, design) [8 marks].

a. $|A_v| = g_m R_C = \dfrac{I_C}{V_T} R_C = \dfrac{I_C R_C}{V_T} = \dfrac{V_{R_C}}{V_T}$. The individual values of I_C and R_C do not matter, only their product.

b. $V_{R_C} = 12 - 6 = 6$ V, so $|A_v|_{max} = 6/0.026 = \mathbf{231}$. More gain needs a larger V_{R_C}, which leaves less V_{CE} and less swing. That is the basic gain-versus-swing trade-off.

c. $V_{R_C} = 150(0.026) = 3.9$ V, so $R_C = 3.9\text{ V}/1\text{ mA} = \mathbf{3.9\ k\Omega}$ (a standard value). $V_{CEQ} = 12 - 3.9 = \mathbf{8.1\ V}$. $I_B = 10\ \mu$A, so $R_B = (12 - 0.7)/10\ \mu\text{A} = \mathbf{1.13\ M\Omega}$.

d. $I_E = 1.01$ mA, so $r_e = 25.74\ \Omega$ and $-R_C/r_e = \mathbf{-151.5}$ (the α difference). With $r_o = 100$ kΩ: $R_C \parallel r_o = 3.75$ kΩ and $A_v = \mathbf{-145.8}$. That is 3 % short of the target. To compensate, raise R_C slightly (about 4.02 kΩ, nearest standard 3.9 kΩ or 4.3 kΩ).

> Also note that fixed bias makes I_C proportional to β, so a real part with a different β shifts both the Q-point and the gain. This is why the divider bias is preferred.

6. Quick recall check

Question

1. $g_m = I_C/V_T$; $r_\pi = \beta/g_m = V_T/I_B$; $r_e = V_T/I_E \approx 1/g_m$; $r_o = V_A/I_C = 1/h_{oe}$.
2. R_1 goes base to ground (in parallel with R_2), R_C goes collector to ground, and R_E is shorted, so it disappears.
3. $v_\pi = i_b r_\pi$ and $g_m r_\pi = \beta$, so $g_m v_\pi = \beta i_b$.
4. $Z_i = R_1 \parallel R_2 \parallel \beta r_e$; $Z_o = R_C \parallel r_o$; $A_v = -(R_C \parallel r_o)/r_e$, with a 180° shift.
5. $|A_v| \approx V_{R_C}/V_T = 5/0.026 \approx 192$, inverting.